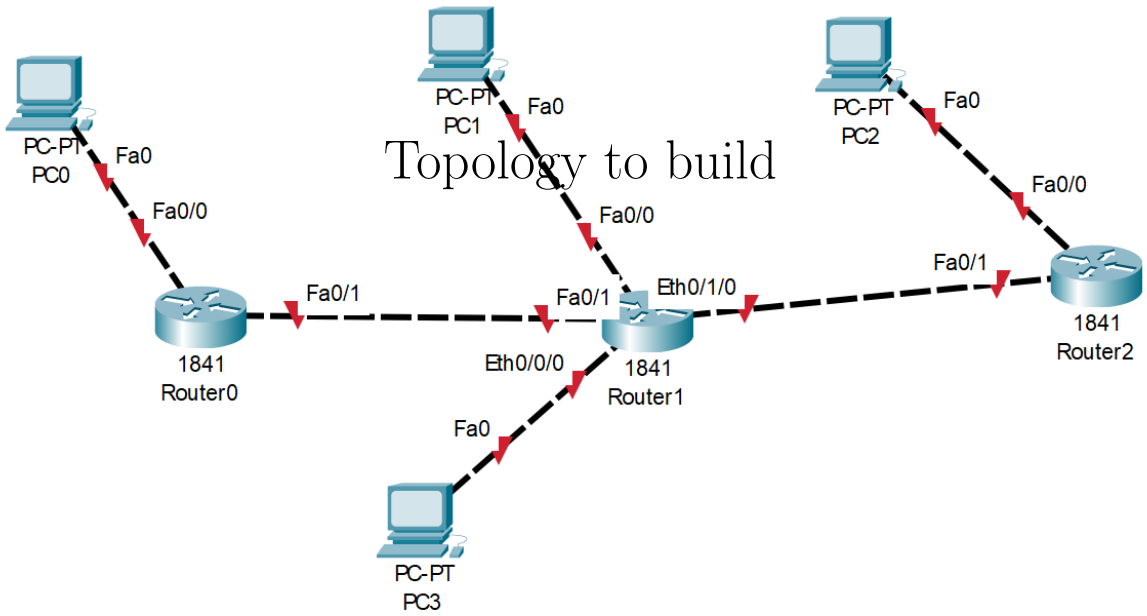
EBU 5403 lab script – online version

In this lab you have ninety minutes to create a topology within Packet Tracer, answer some questions about it and to post the word document **and the .pkt file from saving the topology** to QMPlus. Please do not leave it until the last minute to upload. The system will let you upload several times. To save a topology as .pkt use the save as feature in packet tracer. If you do not include a pkt file with your submission or the pkt file does not match the answers on your script you will lose marks.

- You may not communicate electronically with other students during the test.
- Your test answers are unique. If you copy answers from another student they will be wrong.
- Remember to save your work often as computers can crash. You will not be given extra time if your computer crashes and you did not save your work.
- Emailed answers will not be accepted you must submit through QMPlus.
- Most of the marks for this test are in part one. Part two is harder and for extra marks.



Please notice that some parts are labelled "Part 2". You should not configure those yet.

Device	Interface	IP address	Netmask
PC0	Fa0	131.44.38.13	255.255.0.0
PC0	Gateway	131.44.38.14	255.255.0.0
PC1	Fa0	193.35.16.15	255.255.255.0
PC1	Gateway	193.35.16.16	255.255.255.0
PC3	Fa0	130.35.60.41	255.255.0.0
PC3	Gateway	130.35.60.42	255.255.0.0
Router0	Fa0/0	131.44.38.14	255.255.0.0
Router0	Fa0/1	195.60.59.29	255.255.255.0
Router1	Fa0/0	193.35.16.16	255.255.255.0
Router1	Fa0/1	195.60.59.30	255.255.255.0
Router1	Eth0/0/0	130.35.60.42	255.255.0.0
Router1	Eth0/1/0	Part 2	Part 2
Router2	Fa0/1	Part 2	Part 2
Router2	Fa0/0	Part 2	Part 2
PC2	Fa0	Part 2	Part 2
PC2	Gateway	Part 2	Part 2

RIP for Router0 should include 131.44.0.0, 195.60.59.0. RIP for Router1 should include 195.60.59.0, 193.35.16.0, 130.35.0.0. RIP for Router2 should be set in part 2.

Part ONE

When asked to take a picture please use a snapshot facility on your computer not a mobile phone picture.

When asked to Cut and paste TEXT from commands you should copy the text including the command and its response. An example of what you should cut and paste is shown below.

```
PC>tracert 12.12.12.12

Tracing route to 12.12.12.12 over a maximum of 30 hops:

  1   100 ms  100 ms    100 ms      10.10.10.10
  2   100 ms  100 ms    100 ms      12.12.12.12

Trace complete.
```

Create the topology shown using 1841 routers and also the W1C-1ENET modules. Make sure you connect the correct interfaces. It will help if you set options-preferences-Always show port labels. You should use the wire type shown in the diagram. Add all IP addresses and gateways given. Configure RIP as described. Turn on interfaces. All lights should be green between PC0, PC1, PC3, Router0, Router 1 (except Eth0/1/0). Other lights should be red.

i) Use the windows snapshot program to capture a picture of the topology with these lights green. Paste it into your report.

(6 marks)

ii) On PC0 ping 131.44.38.14 (router 0). It should return a correct response. Cut and paste a picture of the ping command into your report.

(2 marks)

iii) On PC0 ping 195.60.59.29 (PC1). It should return a correct response. Cut and paste a picture of the ping command into your report.

(2 marks)

iv) On PC1 ping 193.35.16.16 (router1). It should return a correct response. Cut and paste a picture of the ping command into your report.

(2 marks)

v) On PC3 ping 131.44.38.13 (PC 0). It should return a correct response. Cut and paste a picture of the ping command into your report.

(2 marks)

vi) On PC0 tracert 131.44.38.13 (PC1). It should return a number of correct responses. Cut and paste the TEXT (not a picture) into your report.

(3 marks)

vii) On PC3 tracert 131.44.38.13 (PC0). It should return a number of correct responses. Cut and paste the TEXT (not a picture) into your report.

(3 marks)

Suggestion: Why not submit the word document **and .pkt file** to QMPlus before moving to part two? You can update them later.

Part TWO

Now complete the system by assigning IP addresses for the remaining part of the system that were labelled "Part 2". You are not given IP addresses for these. You must use your knowledge of subnetworks to work out some values for these.

You will also need to add to the RIP table for router 1 and router 2. If you have done it correctly all lights should be green now.

i) Use the windows snapshot program to capture a picture of the topology now. Paste it into your report.

(2 marks)

ii) Put into your report the new IP addresses and netmasks you used for each router making clear which interface is associated with each.

(1 mark)

iv) On PC2 ping 131.44.38.13 (PC0). It should return a number of correct responses. Cut and paste a picture of the ping command into your report.

(1 marks)

iv) On PC2 tracert 131.44.38.13 (PC0). It should return a number of correct responses. Cut and paste the TEXT (not a picture) into your report.

(1 marks)

Do not forget to submit the word document **and .pkt file** to QMPlus.